

Correction - Calculs sur les fractions

Exercice n°1

a. $(\frac{5}{4} - \frac{7}{2})(\frac{2}{3} + \frac{1}{6})$

$$= (\frac{5}{4} - \frac{7 \times 2}{2 \times 2})(\frac{2 \times 2}{3 \times 2} + \frac{1}{6})$$

$$= (\frac{5}{4} - \frac{14}{4})(\frac{4}{6} + \frac{1}{6})$$

$$= \frac{-9}{4} \times \frac{5}{6}$$

$$= \frac{-45}{24}$$

$$= \frac{-3 \times 3 \times 5}{2 \times 2 \times 2 \times 3}$$

$$= \frac{-15}{8}$$

b. $(\frac{1}{4} - \frac{1}{5}) \times (7 + \frac{37}{9})$

$$= (\frac{1 \times 5}{4 \times 5} - \frac{1 \times 4}{5 \times 4}) \times (\frac{7 \times 9}{1 \times 9} + \frac{37}{9})$$

$$= (\frac{5}{20} - \frac{4}{20}) \times (\frac{63}{9} + \frac{37}{9})$$

$$= \frac{1}{20} \times \frac{100}{9}$$

$$= \frac{100}{180}$$

$$= \frac{2 \times 2 \times 5 \times 5}{2 \times 2 \times 3 \times 3 \times 5}$$

$$= \frac{5}{9}$$

Exercice n°2

a. $\frac{7}{3} \times \frac{5}{4} + \frac{1}{6}$

$$= \frac{35}{12} + \frac{1 \times 2}{6 \times 2}$$

$$= \frac{35}{12} + \frac{2}{12}$$

$$= \frac{37}{12}$$

b. $\frac{9}{28} \times \frac{7}{5} + \frac{10}{3} \times \frac{6}{25}$

$$= \frac{9}{20} + \frac{4 \times 4}{5 \times 4}$$

$$= \frac{9}{20} + \frac{16}{20}$$

$$= \frac{25}{20}$$

$$= \frac{5 \times 5}{2 \times 2 \times 5}$$

$$= \frac{5}{4}$$

Exercice n°3

$$\frac{\frac{4}{3} - \frac{5}{6}}{\frac{1}{2} + 1}$$

a.

$$= \frac{\frac{4 \times 2}{3 \times 2} - \frac{5}{6}}{\frac{1}{2} + \frac{1 \times 2}{1 \times 2}}$$

$$= \frac{\frac{8}{6} - \frac{5}{6}}{\frac{1}{2} + \frac{2}{2}}$$

$$= \frac{\frac{3}{6}}{\frac{3}{2}}$$

$$= \frac{3}{6} \times \frac{2}{3}$$

$$= \frac{2}{3} \times \frac{3}{3}$$

$$= \frac{2}{3}$$

$$= \frac{3}{6} \times \frac{2}{3}$$

$$= \frac{3 \times 2}{2 \times 3 \times 3}$$

$$= \frac{1}{3}$$

$$\text{b. } 2 - \frac{5}{12}$$

$$\frac{1}{3} - \frac{8}{5}$$

$$= \frac{2 \times 12}{1 \times 12} - \frac{5}{12}$$

$$= \frac{1 \times 5}{3 \times 5} - \frac{8 \times 3}{5 \times 3}$$

$$= \frac{24}{12} - \frac{5}{12}$$

$$= \frac{5}{15} - \frac{24}{15}$$

$$= \frac{19}{12}$$

$$= \frac{-19}{15}$$

$$= \frac{19}{12} \times \frac{-15}{19}$$

$$= \frac{19 \times (-3) \times 5}{2 \times 2 \times 3 \times 19}$$

$$= \frac{-15}{12}$$

$$= \frac{-3 \times 5}{2 \times 2 \times 3}$$

$$= \frac{-5}{4}$$

Exercice n°4

$$\begin{aligned}
 & \frac{\frac{4}{3} - 1}{\frac{7}{6} - 2} \\
 \text{a. } & \frac{\frac{4}{3} - \frac{1 \times 3}{1 \times 3}}{\frac{7}{6} - \frac{2 \times 6}{1 \times 6}} \\
 & = \frac{\frac{4}{3} - \frac{3}{3}}{\frac{7}{6} - \frac{12}{6}} \\
 & = \frac{\frac{4}{3} - \frac{3}{3}}{\frac{7}{6} - \frac{12}{6}} \\
 & = \frac{\frac{1}{3}}{\frac{-5}{6}} \\
 & = \frac{1}{3} \times \frac{-6}{5} \\
 & = \frac{-2 \times 3}{3 \times 5} \\
 & = \frac{-2}{5} \\
 & \frac{\frac{2}{3} + \frac{1}{2}}{\frac{17}{9} - \frac{1}{3}} \\
 \text{b. } & \frac{\frac{2 \times 2}{3 \times 2} + \frac{1 \times 3}{2 \times 3}}{\frac{17}{9} - \frac{1 \times 3}{3 \times 3}} \\
 & = \frac{\frac{4}{6} + \frac{3}{6}}{\frac{17}{9} - \frac{3}{9}} \\
 & = \frac{\frac{7}{6}}{\frac{14}{9}}
 \end{aligned}$$

$$\begin{aligned}
 & \frac{7}{6} \\
 &= \frac{\frac{7}{6}}{\frac{14}{9}} \\
 &= \frac{7}{6} \times \frac{9}{14} \\
 &= \frac{7 \times 3 \times 3}{2 \times 3 \times 2 \times 7} \\
 &= \frac{3}{4}
 \end{aligned}$$

Exercice n°5

$$\begin{aligned}
 \mathbf{C} &= \frac{22}{33} - \frac{35}{18} \times \frac{15}{7} \\
 &= \frac{2 \times 11}{3 \times 11} - \frac{5 \times 7 \times 3 \times 5}{2 \times 3 \times 3 \times 7} \\
 &= \frac{2}{3} - \frac{25}{6} \\
 &= \frac{2 \times 2}{3 \times 2} - \frac{25}{6} \\
 &= \frac{4}{6} - \frac{25}{6} \\
 &= \frac{-21}{6} \\
 &= \frac{-3 \times 7}{2 \times 3} \\
 &= \frac{-7}{2} \\
 \mathbf{A} &= \frac{1 + \frac{2}{3}}{1 - \frac{2}{3}} \\
 &= \frac{\frac{1 \times 3}{1 \times 3} + \frac{2}{3}}{\frac{1 \times 3}{1 \times 3} - \frac{2}{3}}
 \end{aligned}$$

$$\begin{aligned}
 & \frac{3}{3} + \frac{2}{3} \\
 = & \frac{\frac{3}{3} - \frac{2}{3}}{\frac{5}{3}} \\
 = & \frac{\frac{1}{3}}{\frac{5}{3}} \\
 = & \frac{5}{3} \times \frac{3}{1} \\
 = & \frac{5 \times 3}{3} \\
 = & 5
 \end{aligned}$$

$$T = -6 - \frac{5}{2}$$

$$= -6 - \frac{5}{3 \times 5} \times \frac{1}{2}$$

$$= -6 - \frac{1}{6}$$

$$= \frac{-6 \times 6}{1 \times 6} - \frac{1}{6}$$

$$= \frac{-36}{6} - \frac{1}{6}$$

$$= \frac{-37}{6}$$